



Sanjivani Rural Education Society's
Sanjivani College of Engineering, Kopargaon-423603
Department of Computer Engineering

A Presentation On
Mini Project – Project-Based Learning (PBL)
Subject: Applied Data Science Laboratory (PCCO314A)
“SmartStock Intelligent Inventory Optimization Platform”
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Problem definition

- **Market Complexity:** Footwear inventory is hard to manage due to shifting trends, regional differences, and seasonal demand.
- **The Cost of Overstocking:** Excess inventory traps working capital, raises warehousing costs, and risks leaving products unsold.
- **The Cost of Stockouts:** Insufficient inventory causes missed sales opportunities and damages trust with retailers.
- **Inefficient Processes:** Traditional manual methods fail to adapt to real-time market variations, leading to unreliable business decisions.



Objectives

- **Predictive Forecasting:** Forecast daily unit sales for specific product and brand combinations.
- **Automated Replenishment:** Dynamically calculate Safety Stock and Reorder Points using data-driven logic.
- **Stock Health Management:** Automatically identify slow-moving items and liquidate deadstock.
- **Actionable Dashboards:** Deliver a web-based dashboard that provides automated daily alerts for orders and discounts.



Dataset Description

Attribute	Details
Dataset Name	FootWare_Wholesale_Sales_Dataset.csv
Total Records	11,115 wholesale transactions
Original Features	16 columns
Project Goal	Forecast footwear wholesale demand
Target Variable	Quantity Sold
Machine Learning Models	Random Forest, XGBoost

Dropped Features (To Prevent Data Leakage)

- Total Revenue, Profit, Margin, Tax-related columns
- Dealer Location

Key Features Used for Forecasting

- **Date** – Captures seasonality and sales trends
- **Product & Brand** – Represent product popularity
- **Size** – Identifies high-demand shoe sizes
- **Dealer** – Reflects dealer purchasing patterns
- **Stock Availability** – Ensures sales depend on inventory
- **Unit Price (₹)** – Captures price-demand relationship



Feature Engineering and Preprocessing

1. Temporal Features

Derived from the **Date** column:

- Day_of_Week
- Day_of_Month
- Month
- Quarter
- Year
- Week_of_Year
- Is_Weekend
- Is_Month_Start
- Is_Month_End

2. Lag Features

Historical demand values:

- Lag_1
- Lag_7
- Lag_14
- Lag_30

3. Rolling Features

Recent demand trends, volatility:

- Rolling_7_Mean
- Rolling_14_Mean
- Rolling_30_Mean
- Rolling_7_Std

4. Categorical Encoding

- Product_Encoded
- Brand_Encoded
- Dealer_Encoded

Preprocessing Steps:

Data Cleaning: Converted Date and percentage columns into numeric format.

Feature Selection: Removed leakage and redundant columns.

Label Encoding: Converted Product, Brand, and Dealer into numeric values.

Standardization: Applied StandardScaler to numerical features.

Time Split: Used 75% data for training and 25% for testing.

Final Output: Created 23 optimized features

Feature engineering transforms raw data into meaningful numerical features that help machine learning models capture seasonality, trends, and historical demand patterns.



Model Development

- **Candidate Models:** Linear Regression, Random Forest, and XGBoost were selected to compare baseline, ensemble, and advanced boosting approaches.
- **Hyperparameter Tuning:** RandomizedSearchCV was used to optimize model parameters and improve forecasting accuracy.
- **Validation Strategy:** A chronological 75%–25% train-test split and 5-fold cross-validation ensured robust and realistic evaluation.
- **Performance Metrics:** MAE, RMSE, R^2 , and MAPE were used to measure prediction accuracy and error behavior.
- **Best Model Selection:** The model with the highest Test R^2 score was automatically selected as the final model.
- **Model Output:** The winning model was saved as best_model.pkl along with preprocessing components for deployment.

System Architecture

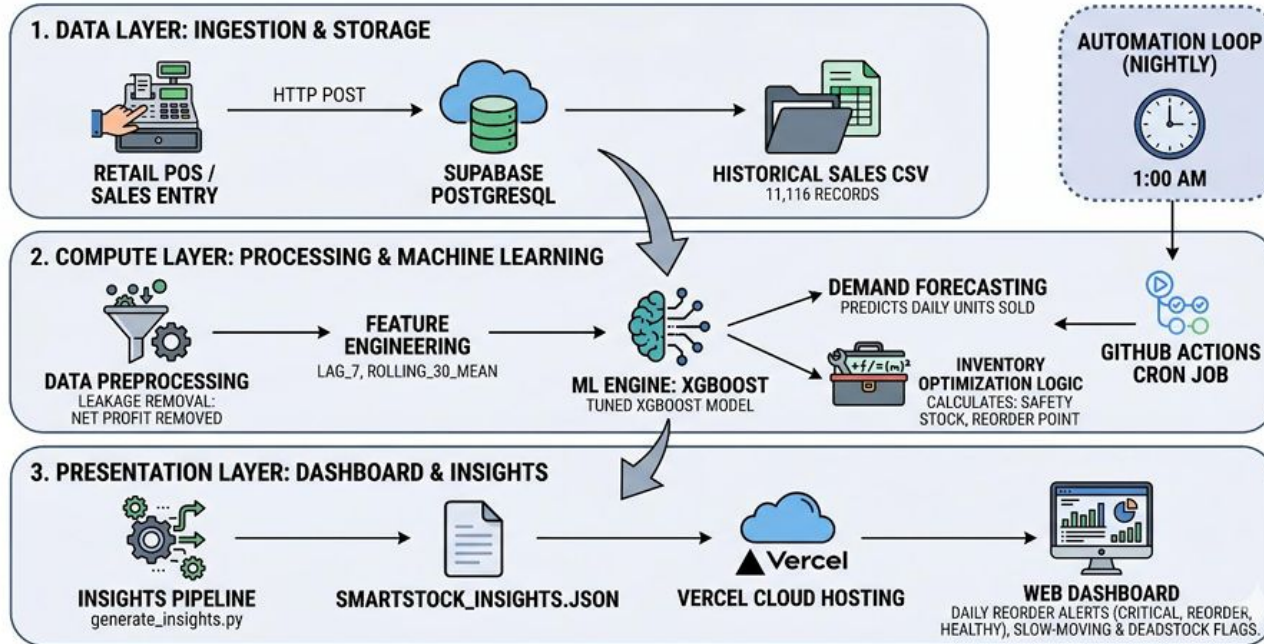


Fig.: System Architecture And Working



Result analysis

Metric	Value	Interpretation
Best Model	XGBoost (Tuned)	Highest forecasting accuracy
R ² Score	0.8451	Explains 84.5% of demand variation
MAE	5.56 units	Average prediction error
MAPE	11.4%	Low percentage forecasting error

Business Impact: Reduced stockouts and overstocking, identified deadstock, improved cash flow, and enabled data-driven inventory decisions with 84.5% forecasting accuracy.

Key Demand Drivers

- Rolling 7-Day Average
- Is_Weekend
- Rolling 14-Day Average
- Day_of_Week
- Lag_7

Inventory Insights

- Calculates **Safety Stock** and **Reorder Points**
- Classifies inventory into **Fast-Moving, Normal, Slow-Moving, and Deadstock**
- Generates automatic **Restock Recommendations**



Conclusion

- **Demand Forecasting:** XGBoost accurately predicts footwear demand with $R^2 = 0.845$, explaining 84.5% of sales variation.
- **Inventory Optimization:** Dynamic **Safety Stock**, **Reorder Point (ROP)**, and **EOQ** minimize stockouts and excess inventory.
- **Stock Health Analysis:** Inventory is classified into **Fast-Moving**, **Normal**, **Slow-Moving**, and **Deadstock** to improve capital utilization.
- **Decision Support Dashboard:** Provides real-time alerts and automatic restocking recommendations.
- **Business Outcome:** Improved forecasting accuracy, reduced waste, optimized inventory investment, and enhanced supply chain efficiency.



References

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- **Data Science Framework:** IBM, "CRISP-DM Methodology," 2024. <https://www.ibm.com/topics/crisp-dm>
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- **Machine Learning Library:** scikit-learn Developers, "scikit-learn Documentation," 2024. <https://scikit-learn.org/stable/documentation.html>
- **Project Dataset Source:** Kaggle, "Retail Sales Forecasting Dataset," 2023. <https://www.kaggle.com/datasets/rohitsahoo/sales-forecasting>



THANK YOU!!
